Solution Requirements

Date	23 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	3 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users securely log in to access the HDI prediction system.	High
2	Authorization Levels	Admin manages datasets/models; users perform predictions and view reports.	High
3	External Interfaces	Uses CSV datasets and Scikit-learn, Pandas, NumPy, Flask.	High
4	Transactions Processing	Validates input and generates HDI predictions.	High
5	Reporting	Displays HDI category, confidence score, charts and downloadable reports.	High

6	Business Rules	Prediction uses Life Expectancy, Mean Years of Schooling, Expected Years of Schooling and GNI per Capita.	High
7	Compliance	Follows data privacy and ethical AI practices.	Medium
8	Other	Supports model retraining and prediction history.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	Fast prediction processing.	Prediction within 2 seconds
2	Scalability	Handle increasing users and datasets.	1000+ concurrent users
3	Security & Data Privacy	Secure login and encrypted communication.	HTTPS & encrypted storage
4	Reliability & Availability	Stable system operation.	99.9% uptime
5	Usability & Accessibility	Responsive and user-friendly interface.	Works on desktop/mobile
6	Other	Maintainability and portability.	Modular documented code